

LLD and HLD Roadmap

LLD Topics

S.O.L.I.D Principles
Strategy Pattern
Observer Pattern
Design Notify-Me Button Functionality
Decorator Pattern
Design Pizza Billing System
Factory Pattern
Design Parking Lot
Abstract Factory Pattern
Design Snake n Ladder game
Chain of Responsibility Pattern Design WhatsApp
Design Elevator System
Proxy Pattern
Design Car Rental System
Null Object Pattern
Design Logging System
State Pattern
Design Tic-Tac-Toe game
Composite Pattern
Design BookMyShow & Concurrency handling
Adapter Pattern
Design Vending Machine
Singleton Pattern
Design ATM
Builder Pattern
Design Chess game
Prototype Pattern
Design File System
Bridge Pattern
Design Splitwise
Façade Pattern
Splitwise Simplify Algorithm / Optimal Accounting Balancing
Flyweight Pattern
Design CricBuzz / CricketInfo
Command Pattern
Design True Caller
Interpreter Pattern
Design Car Booking Service like Ola, Uber
Iterator Pattern

HLD topics

Learn About Network Protocols (TCP, Websocket, HTTP etc.)
Client-Server Vs Peer 2 Peer Architecture
C.A.P Theorem
Microservices Imp. Design Patterns (SAGA pattern, Strangler Pattern)
Scale from 0 to Million Users
Design Consistent Hashing
Design URL Shortening
Back of the Envelope Estimation
Design Key-Value Store
SQL vs NoSQL, When to Use Which DB
Design Rate Limiter
Design Search Autocomplete System / Typeahead System
Understand Message Queue , Kafka etc.
What is Proxy Servers
What is CDN
Storage types:
(Block Storage, File Storage, Object Storage (S3) , RAID)
File System
(Google File System, HDFS)
Bloom Filter
Merkle Tree , Gossiping Protocol
Caching
(Cache Invalidation, Cache eviction)
How to Scale Database
Sharding (Horizontal and Vertical)
Partitioning
Replication, Mirroring
Leader Election
Indexing etc.
Design Notification System
Design Pastebin
Design Twitter
Design Dropbox
Design Instagram
Design YouTube
Design Google Drive
Design Web Crawler
Design Facebook News Feed / Newsfeed System

LLD Topics

Design Online Hotel Booking System
Mediator Pattern
Design Library Management System
Memento Pattern
Design Traffic Light System
Template Method Pattern
Design Meeting Scheduler
Visitor Pattern
Design Online Voting System
Design Inventory Management System
Design Cache Mechanism
Design LinkedIn
Design Amazon
Design Airline Management System
Design Stock Exchange System
Design Learning Management System
Design a Calendar Application
Design (LLD) Payment System
Design (LLD) Chat based system
Design Food delivery app like Swiggy and Zomato
Design Community Discussion Platform
Design Restaurant Management System
Design Bowling Alley Machine
Design (LLD) Rate Limiter

HLD topics

Design Ticket Master
Design NearByFriends or Yelp